

1. Administrative & Study Identification

Full Study Title: A Study Evaluating the Efficacy and Safety of Inavolisib Plus CDK4/6 Inhibitor and Letrozole vs Placebo + CDK4/6i and Letrozole in Participants With Endocrine-Sensitive PIK3CA-Mutated, Hormone Receptor-Positive, HER2-Negative Advanced Breast Cancer **Short Title/Acronym:** INAVO123 **Protocol Number:** WO45654 **NCT Number:** NCT06790693 **Trial Status:** Recruiting **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Inavolisib (Itovebi) / Placebo + CDK4/6 Inhibitor (e.g., Palbociclib) + Letrozole **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of inavolisib combined with a CDK4/6 inhibitor (CDK4/6i) and letrozole compared to placebo plus a CDK4/6i and letrozole in terms of Progression-Free Survival (PFS). **Secondary Objectives:** To evaluate Overall Survival (OS), Investigator-assessed Objective Response Rate (ORR), Investigator-assessed Duration of Response (DOR), Clinical Benefit Rate (CBR), and Patient-reported outcomes (including Time to Confirmed Deterioration (TTCD) in Pain and treatment side-effect bother).

2.2 Background & Rationale To address the need for improved first-line treatments for patients with endocrine-sensitive PIK3CA-mutated hormone receptor-positive (HR+), human epidermal growth factor receptor 2-negative (HER2-) advanced breast cancer (ABC). While current standard-of-care includes a CDK4/6 inhibitor plus endocrine therapy, adding an oral, targeted PI3K inhibitor (inavolisib) aims to provide well-tolerated, durable disease control, significantly delay disease progression, and improve overall survival for patients who often have a poor prognosis.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 Randomized (Inavolisib arm vs. Placebo arm)

3.2 Planned Enrollment & Duration Trial Status: Recruiting (Currently recruiting participants) **Target N\$:** 450 participants **Number of Sites:** ~219 locations globally **Estimated Study Start:** 09-Apr-2025 **Estimated Primary Completion:** 30-May-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Women or men ≥ 18 years with histologically or cytologically confirmed carcinoma of the breast. 2. Documented ER-positive and/or progesterone receptor-positive tumor according to American Society of Clinical Oncology/College of American Pathologists (ASCO/CAP) guidelines. 3. Documented HER2-negative tumor according to ASCO/CAP guidelines. 4. De-novo HR+, HER2- ABC, or, alternatively, relapsed HR+, HER2- ABC after at least 2 years of standard neoadjuvant/adjuvant endocrine therapy without disease progression during that treatment and disease-free interval of at least 1 year since the completion of that treatment. 5. Participants who have bilateral breast cancers which are both HR-positive and HER2-negative. 6. Confirmation of biomarker eligibility (qualifying PIK3CA mutation determined by central ctDNA or pre-existing local tissue). 7. Consent to provide fresh or archival tumor tissue specimen. 8. Measurable disease per Response Evaluation Criteria in Solid Tumors, Version 1.1 (RECIST v1.1). 9. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 10. Adequate hematologic and organ function within 14 days prior to initiation of study treatment.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intention of becoming pregnant during the study or within the time frame in which contraception is required. 2. Metaplastic breast cancer. 3. Any prior systemic therapy for locally advanced unresectable or metastatic breast cancer. 4. Type 2 diabetes requiring ongoing systemic treatment at the time of study entry; or any history of Type 1 diabetes. 5. Any history of leptomeningeal disease or carcinomatous meningitis. 6. Known and untreated, or active CNS metastases. Participants with a history of treated CNS metastases are eligible. 7. Active inflammatory or infectious conditions in either eye or history of idiopathic or autoimmune-associated uveitis in either eye. 8. Symptomatic active lung disease. 9. History of or active inflammatory bowel disease; or any active bowel inflammation. 10. Prior hematopoietic stem cell or bone marrow transplantation. 11. Treatment with strong cytochrome P450 (CYP) 3A4 inhibitors or strong CYP3A4 inducers within 4 weeks or 5 drug-elimination half-lives, prior to initiation of study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inavolisib (9 mg daily) OR Placebo + a CDK4/6i (e.g., Palbociclib daily for 21 days on, 7 days off) + Letrozole (Trade Name: Femara; ATC Code: L02BG04; 2.5 mg daily) administered in 28-day cycles. Pre- and perimenopausal women, as well as men, will also receive a luteinizing hormone-releasing hormone (LHRH) agonist. **Route of Administration:** Oral **Duration of Treatment:** Until disease

progression, unacceptable toxicity, patient withdrawal of consent, death, or study termination.

5.2 Concomitant & Prohibited Therapy Permitted: LHRH agonists (required for pre- and perimenopausal women and men), and standard supportive care for breast cancer symptoms. **Prohibited:** Treatment with strong cytochrome P450 (CYP) 3A4 inhibitors or strong CYP3A4 inducers within 4 weeks (or 5 drug-elimination half-lives) prior to initiation of study treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Biomarker confirmation (PIK3CA), Tumor Imaging (RECIST 1.1)
2	0	Randomization (1:1), First Dose Administration
3-18	Every 28 Days	Treatment dispensing, Safety Labs (Glucose, LFTs), Efficacy Assessment, Patient-Reported Outcomes (PROs)
19-72	Up to 7 Years	Disease Progression confirmation, Final Vital Signs, Exit Interview, IP Accountability, Overall Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) - Time from randomization to the first occurrence of disease progression or death from any cause, whichever occurs first (up to 7 years). **Measurement Method:** Investigator-assessed via Response Evaluation Criteria in Solid Tumors (RECIST) v1.1, utilizing a stratified log-rank test.

7.2 Secondary & Exploratory Endpoints

1. Overall Survival (OS) - Time from randomization to death from any cause.
2. Investigator-assessed Objective Response Rate (ORR).
3. Investigator-assessed Duration of Response (DOR).
4. Investigator-assessed Clinical Benefit Rate (CBR).
5. Time to Confirmed Deterioration (TTCD) in Pain and patient-reported treatment side-effect bother.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine laboratory monitoring (including fasting glucose, HbA1c, and complete blood counts) and AE grading according to NCI CTCAE criteria. **Serious Adverse Events (SAEs):** Standard 24-hour expedited reporting to the Sponsor and local regulatory authorities. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) will oversee safety and interim efficacy.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population will be used for primary efficacy endpoint analysis. **Sample Size**

Justification: $N=450$ participants to adequately power the PFS and OS analyses. The primary endpoint will be assessed via a stratified log-rank test with a two-sided significance level of $\alpha = 0.05$. **Handling of Missing Data:** Right-censoring will be applied to participants who have not reached the event of interest (progression or death) at the time of the data cut-off, employing Kaplan-Meier methods.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard risk-based onsite and remote monitoring. **Audit Trail:** Validated Electronic Data Capture (EDC) systems will be utilized for eCRF entry, locking, and generation of data clarification forms.

11. Study Identifiers & Governance

Primary ID: W045654 **Registry Number:** NCT06790693 **Posting ID:** 652551985424183916 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** INAVO123 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Inavolisib Plus a CDK4/6 Inhibitor and Letrozole Versus Placebo Plus a CDK4/6 Inhibitor and Letrozole in Patients With Endocrine-Sensitive PIK3CA-Mutated, Hormone Receptor-Positive, HER2-Negative Advanced Breast Cancer

12. Clinical Framework (Oncology Specific)

Disease Area: Breast Cancer **Preferred UMLS Name:** Metastatic Hormone Receptor Positive HER2 Negative Breast Cancer **Disease**

Preferred Name: Malignant neoplasm of breast **Semantic Type:**

Neoplastic Process **Disease Definition:** A primary or metastatic malignant neoplasm involving the breast. The vast majority of cases are carcinomas arising from the breast parenchyma or the nipple. Malignant breast neoplasms occur more frequently in females than in males. **MeSH Code:** D001943 **SNOMED ID:** 254837009

Medical Methodology: Targeted precision oncology combining a PI3K inhibitor with Guideline-Determined Medical Therapy (CDK4/6i + Endocrine therapy) **Optimization Target:** Maximal delay of disease progression and prolonged Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: First-Line standard clinical pathway for ABC integrating biomarker-driven targeted therapy. **Education Protocol (HCP):** Investigator site training prior to site initiation, focusing on dosing, toxicity management (e.g., hyperglycemia, stomatitis), and RECIST 1.1 evaluations. **Education Protocol (Patient):** Informed consent process detailing the risks of investigational products, along with self-learning materials regarding expected side effects. **Quality Audit Mechanism:** Periodic Independent Data Monitoring Committee (IDMC) reviews and central tumor imaging review.

14. Observation & Follow-Up Schedule

Total Duration: Estimated up to 7 Years (84 months) **Onsite Visit Cadence:** Every 28 Days (Day 1 of each 28-day treatment cycle) **Remote Monitoring Frequency:** Intermittent monitoring based on patient status and local site protocols. **Checklist Review Interval:** Routine PRO (Patient-Reported Outcome) questionnaire collection linked with clinical visits.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				